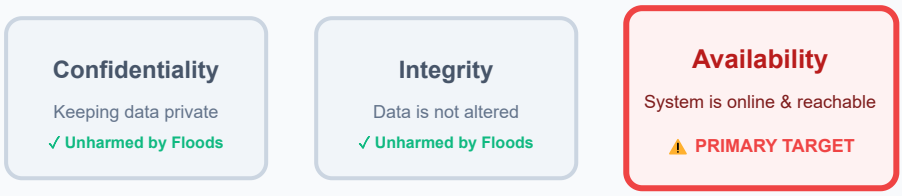
VISUAL DIAGRAMS & CHARTS

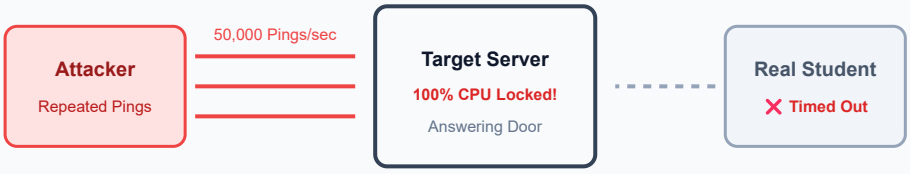
FLOOD ATTACKS AT A GLANCE

A Visual Guide to Network Exhaustion, Traffic Telemetry, and Defense Topologies

1. The CIA Triad: Where Floods Strike



2. Visual Analogy: The Doorbell Flood



The Attack Objective

A flood attack does not compromise secret keys or install malware. Its sole goal is to overwhelm capacity until the system cannot operate.

The Defensive Challenge

Because each individual ping or request appears structurally valid, simple inspection cannot distinguish a malicious packet from a normal one without rate-tracking.

1. How Network Packets Travel from Client to Server



2. Anatomy of a Network Packet Envelope



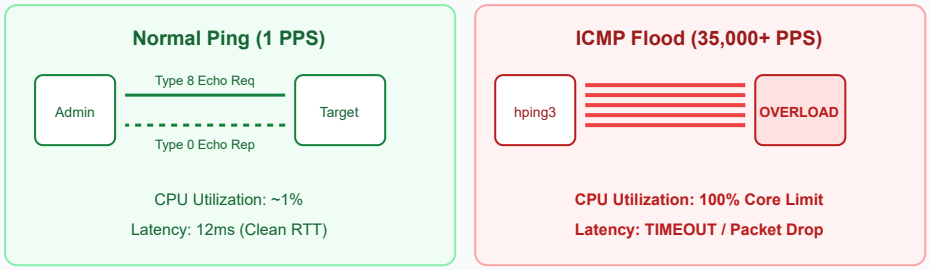
Bandwidth vs Latency

Bandwidth: Highway lane capacity (Gbps).
Latency: Time required to complete the round-trip (ms).

Hardware Packet Interrupts

Every single packet demands attention from the server's CPU kernel. Processing 40,000 headers per second leaves zero processing power for applications.

1. Normal Ping vs ICMP Flood Comparison



2. ICMP Message Code Architecture

ICMP Type 8: Echo Request

Sent by the diagnostic client. Contains identifier, sequence number, and optional data payload. Triggers an obligatory response from the receiver.

ICMP Type 0: Echo Reply

Sent by the target server. Must mirror the exact identifier and data payload of the incoming Echo Request, forcing the server to construct packets in RAM.

Why ICMP Floods are Deadly to Small Servers

Because ICMP is connectionless, the attacker does not need to complete a handshake or maintain memory state. The attacker simply fires raw packets at wire speed. The target, however, must expend CPU cycles decoding each packet, computing checksums, and building reply packets.

1. 4-Quadrant Attack Vector Taxonomy

1. ICMP Flood (Layer 3)
Target: Host CPU & SoftIRQs
Defense: iptables ICMP rate limiting

2. TCP SYN Flood (Layer 4)
Target: TCP Backlog Queue (RAM)
Defense: SYN Cookies & SYN rate limits

3. UDP Flood (Layer 4)
Target: Network Bandwidth (Gbps)
Defense: ISP Scrubbing & Port Filtering

4. HTTP Flood (Layer 7)
Target: App Workers & Database
Defense: WAF, CAPTCHAs, Reverse Proxies

2. Network Layer vs Application Layer Exhaustion

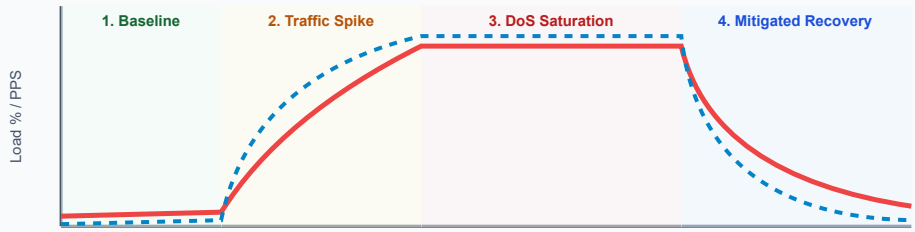
Volumetric (Layer 3 & 4)

Goal is brute force pipe saturation. Measured in Gigabits per second (Gbps) or Millions of Packets per second (Mpps). Requires high attacker bandwidth or botnet amplification.

Application (Layer 7)

Goal is exhausting backend computation (e.g. database search queries or SSL handshakes). Requires very low bandwidth from attacker but brings down multi-core databases.

1. The 4-Phase Overload Progression Timeline



Target CPU Utilization (%) Packets Per Second (PPS)

2. Key Observations Across the Timeline

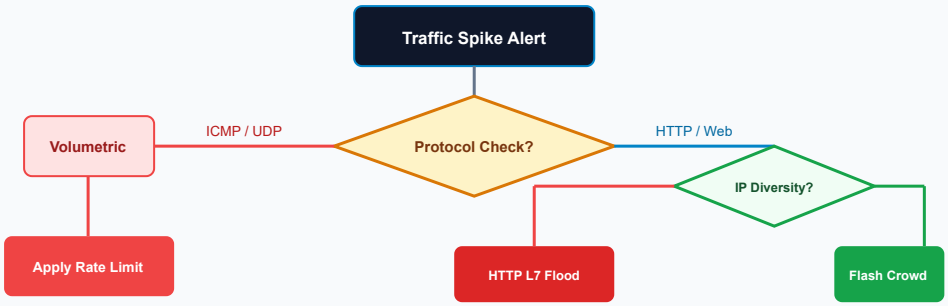
Spike Phase: Immediate Kernel Starvation

Unlike memory leaks which take hours, packet floods exhaust CPU within 2 to 3 seconds of execution as interrupts overwhelm CPU scheduling queues.

Recovery Phase: Kernel Drop Restores CPU

The moment iptables rules are active, packets are dropped at Layer 3 inside Netfilter. CPU returns to healthy baseline almost instantaneously.

1. Incident Response Traffic Spike Decision Tree



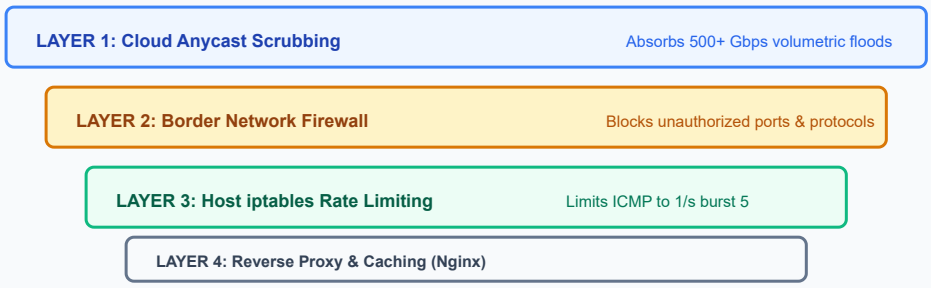
Investigating Packet Headers

Examine whether packets are carrying valid TCP sequence numbers or if they share identical forged payload signatures.

Distinguishing Humans from Bots

Bots fail basic browser JavaScript challenges (e.g., Cloudflare Under Attack mode) and repeat identical URIs thousands of times per minute.

1. Hierarchical Defense Architecture



2. When to Use Which Mitigation Technique

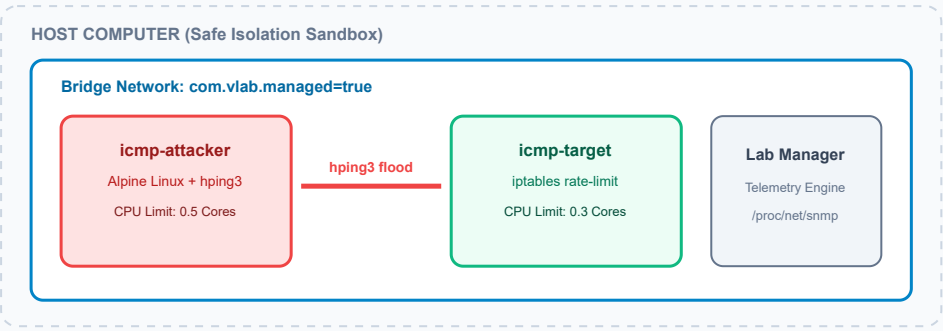
When Local iptables is Sufficient

When attacks are low-to-moderate volume (e.g. single-source DoS in our lab) and network bandwidth is not completely saturated.

When Cloud Scrubbing is Mandatory

When the volume of attack traffic exceeds the physical bandwidth of the host's network cable (e.g. 50 Gbps flood targeting a 1 Gbps port).

1. Isolated Docker Container Network Topology



Isolated Virtual Bridge

No attack packets can leak onto your physical home WiFi or reach internet destinations. Traffic is confined to the Docker bridge.

Resource Throttling

Both containers are capped with CPU and memory limits, ensuring your computer stays completely smooth and responsive during tests.

1. Side-by-Side Dashboard Telemetry Comparison

DURING FLOOD (UNPROTECTED)

```
STATUS:       CRITICAL (DoS)
PACKET RATE:  35,210 PPS
CPU USAGE:    28.4% (Max Capped)
RAM USAGE:    24.2 MB
USER PING:    Request Timed Out ❌
```

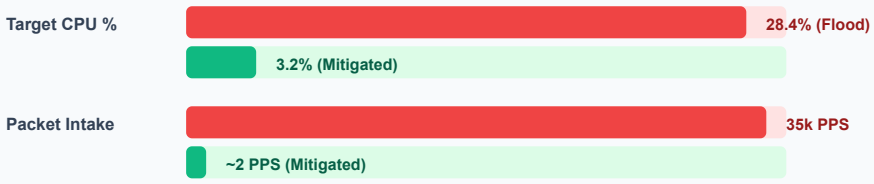
Impact: Target is choked answering Echo Requests. Legitimate users receive 100% packet loss.

AFTER MITIGATION (PROTECTED)

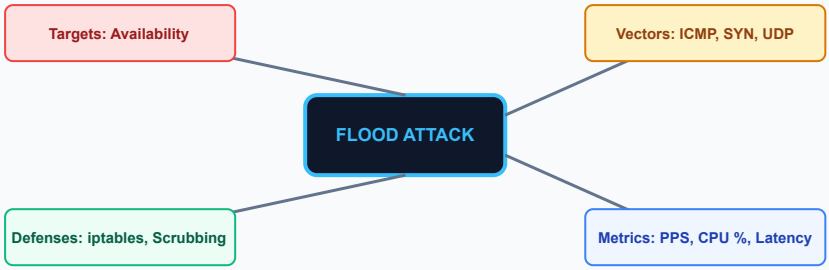
```
STATUS:       HEALTHY
PACKET RATE:  1 to 3 PPS
CPU USAGE:    3.2% (Baseline Idle)
RAM USAGE:    18.5 MB
USER PING:    Reply in 2ms ✅
```

Resolution: iptables rate-limits ICMP to 1/s burst 5. 99.9% of junk is discarded at Layer 3!

2. Visual Bar Metric Comparison



1. Comprehensive Mind Map: Flood Attack Knowledge Base



2. Five Quick Visual Memory Cues for Exams

1. The Doorbell (ICMP)

A flood attack is like ringing a doorbell thousands of times so the staff cannot help real visitors.

2. The Bouncer (Firewall)

A firewall stops unauthorized packets at the door, but alone cannot stop millions of packets flooding the street.

3. The Turnstile (Rate Limiting)

Rate limiting lets 1 person through at a time, keeping the room calm without shutting down the facility.

4. The Highway Bypass (Scrubbing)

Upstream scrubbing centers divert massive multi-gigabit traffic away before narrow residential streets choke.

5. The Golden Cybersecurity Rule

Encryption protects Confidentiality; Defensive Architecture protects Availability. A complete cybersecurity engineer always protects both!